

Correspondence Agent

- Vendor clarification drafts for invoice exceptions
- Current implementation: FastAPI + Streamlit + structured AI output
- Draft-only communication with human review

Purpose

- Creates a professional vendor email when an invoice needs clarification
- Uses verified invoice and exception facts
- Never sends an email automatically
- Never makes the payment approval or rejection decision

When It Runs

- The pipeline receives uploaded documents
- An invoice is successfully extracted
- At least one exception is detected
- The HITL router sends the invoice to human_review
- `pipeline.py` calls `correspondence.draft(record, issues)`

Safe Value Formatting

- `_value()` converts available values to text
- Missing values use a clear fallback instead of invented data
- `_money()` formats numeric values as dollars
- Example: 5900 becomes \$5,900.00
- Invalid values are displayed safely instead of crashing

Issue Details

- `_issue_details()` converts exception codes into readable email lines
- `AMOUNT_MISMATCH` shows PO amount, invoice amount, and difference
- `VENDOR_MISMATCH` shows both vendor values
- `PO_NOT_FOUND` and `RECEIPT_NOT_FOUND` explain missing records
- Duplicate, threshold, and outlier issues have dedicated details

Verified Facts Sent to AI

- `_facts()` prepares a small structured dictionary
- Includes vendor name, invoice number, PO number, and exceptions
- Includes severity, expected value, found value, and amount at risk
- The AI receives verified facts rather than inventing document values

AI Draft Generation

- OpenAI is used only when OPENAI_API_KEY is configured
- temperature=0.2 keeps wording consistent and professional
- VendorEmailDraft requires subject, body, and human approval status
- The prompt forbids changing amounts, references, or exception details
- The email must state that human approval is required

Validation and Fallback

- Reject the AI response if it is missing or does not require approval
- Reject it if the invoice number or PO number is missing
- Any API, network, or parsing failure returns None
- The deterministic template then creates a reliable backup draft

Human Review Workflow

- Streamlit displays the exception evidence and vendor draft
- Reviewer can Approve case, Reject case, or Request clarification
- The decision is recorded through POST /review
- Approval of a case is separate from sending an email
- The current review store is in memory for the running session

Example and Key Message

- Invoice INV-1001, PO-5001, amount at risk \$5,900
- Exception: receipt or proof of delivery was not found
- Correspondence creates a clarification draft for the vendor
- Human review remains the final control
- Key message: AI assists communication, but does not decide or send